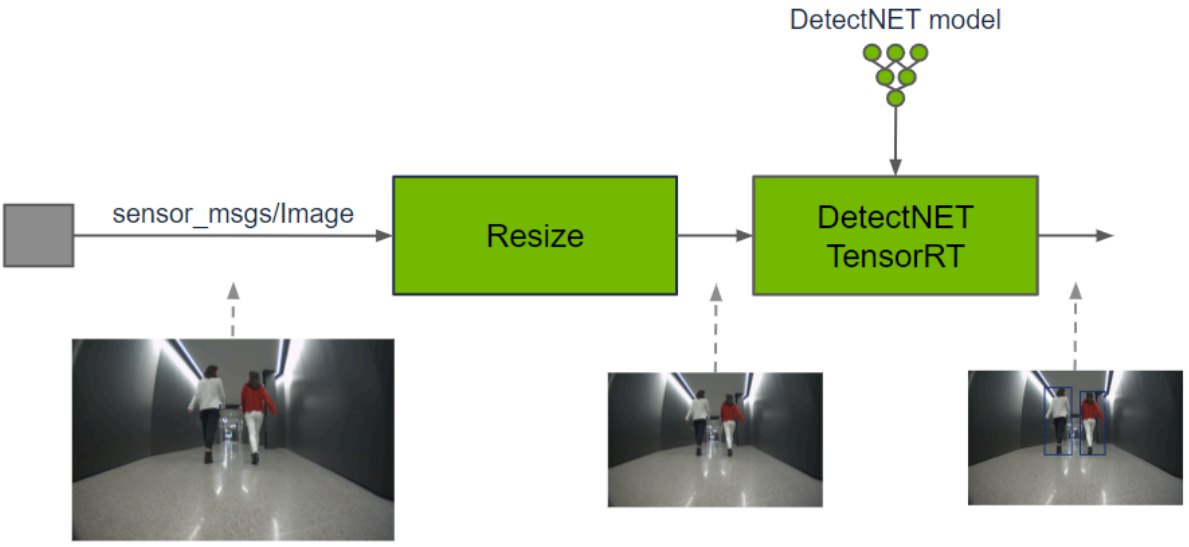
Isaac ROS Object detection

Isaac ROS Object detection
- Overview
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Isaac ROS Object Detection official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_object_detection/index.html

Overview

Isaac ROS provides GPU-accelerated detection models such as RT-DETR, DetectNet and YOLOv8, which predict the object classes and 2D bounding boxes from the input image.



Quick experience

First start the container with the following command

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open the container terminals with `docker exec -it <container name> bash`. Each terminal has to use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: all the terminals below are executed inside the container:

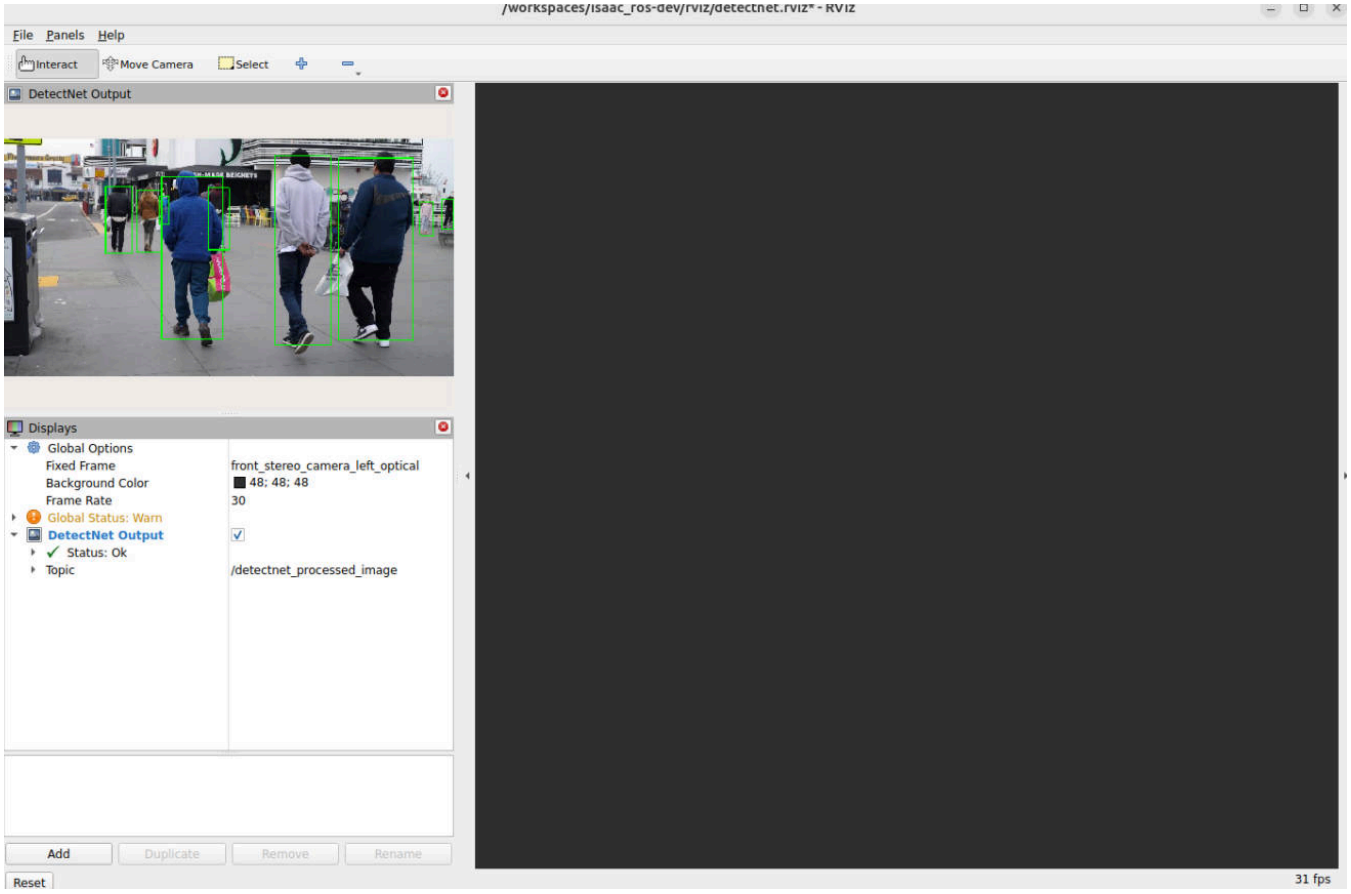
Terminal A: start DetectNet, the synchronized visualization node and RViz:

```
ros2 launch /workspaces/isaac_ros-dev/launch/detectnet_all.launch.py
```

The RViz configuration is `/workspaces/isaac_ros-dev/rviz/detectnet.rviz`, which shows `/detectnet_processed_image`.

Terminal B: play the detection bag in a loop at 0.25 times the speed:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_detectnet/rosbags/detectnet_rosbag \
  --loop --rate 0.25 \
  --remap /image:=/image_rect /camera_info:=/camera_info_rect
```



Terminal C: check the detection and the drawn image output:

```
timeout 8 ros2 topic hz /detectnet/detections
timeout 8 ros2 topic hz /detectnet_processed_image
ros2 topic info -v /detectnet_processed_image
```